

Bhavik Barot

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Summary

Mechanical/Aerospace Engineer (M.Eng., Concordia University) with academic and internship experience in combustion, thermofluid sciences, and structural analysis. Skilled in CAD/FEA tools (CATIA, ANSYS, SolidWorks) and process automation (Python, Power BI). Adaptable and eager to contribute to propulsion, structural, and development teams through Pratt & Whitney Canada's Rotational Program.

Skills

- **Programming & Data Analysis:** Python, MATLAB, Cantera, Power BI, VBA
- **Simulation & CAD Tools:** AutoCAD, Solidworks, CATIA V5, Siemens NX, ANSYS (Fluent, Structural, Classic, APDL), StarCCM+, Nastran, Patran
- **Propulsion & Combustion :** Turbomachinery design, Hydrogen combustion, Combustor design concepts (RQL, Micromix)
- **Mechanical Analysis:** Fatigue (LCF/HCF), Fracture Mechanics, Rotor Dynamics, Thermal Stress on Rotating structures
- **Collaboration & Productivity Tools:** Microsoft Office, Power BI, SharePoint, SAP, Enovia, Jira
- **Languages:** English, French (TEFAQ-B2 Level)

Experience

Pratt & Whitney Canada - Longueuil, Quebec

May 2024 – Sept 2024

Operations Program Management Intern (Full-Time Internship)

- Supported engine program management by developing and enhancing dashboards (Power BI) for production, delivery, and program status reporting.
- Contributed to improvements in data visualization, enabling more accurate tracking of engine program progress and operational performance.
- Assisted in documenting procedures for engine repair and overhaul at intermediate facilities, ensuring compliance with quality standards.

Safran Aerospace - Kirkland, Quebec

March - 2025

Method Designer

- Contributed to the automation of manufacturing instructions using Python and Excel macros, reducing repetitive tasks for engineers.
- Developed a tool using PyCATIA to extract drawing notes and metadata from 2D CAD models, streamlining drawing data transfer into engineering documentation.
- Collaborated with cross-functional teams (design, engineering, and production) to ensure accurate interpretation of Condition of Supply (CSD), BOMs, and 3D/2D models using CATIA V5 and Composer 3DVia.

Vitthalesh Engineering -Gujarat, India

Aug 2021 – Aug 2022

Graduate Engineer – Mechanical Design & Simulation

- Supported design reviews by preparing models and simulations in ANSYS and Siemens NX.
- Performed preliminary fatigue and vibration analysis on mechanical components, assisting senior engineers in validation studies.
- Collaborated with clients and project teams to deliver design documentation aligned with technical requirements.

Education

Concordia University

M.Eng Aerospace Engineering

Sept 2022 – Sept 2024

Major : Combustion (IUT-McGill University), Thermofluid Science

Gujarat Technological University

B.E. Mechanical Engineering

July 2017 – June 2021

Projects

Design Optimization of High-Pressure Turbine and Compressor Blade

Pratt & Whitney Canada

Design of Low Emission Fuel Injection System for Gas Turbine

Siemens Energy Canada

CFD Simulation and Performance Analysis of VKI High Pressure Turbine Blade

Concordia University

Implementation of RQL (Rich Burn Quick Lean) Combustor in Cantera Modeling

McGill University

Research Paper

Conducted research on micromix combustion as a low-NOx strategy for gas turbines by analyzing the aerodynamic behavior of hydrogen flamelets. Focused on flame structure, stabilization, and jet-in-crossflow interactions to assess flame anchoring, temperature uniformity, and NOx reduction. Compared micromix performance with DLN (Dry Low NOx) and RQL (Rich Burn Quick Lean) technologies from an aero-thermal standpoint, highlighting benefits in stability, mixing efficiency, and pressure loss control.